MATH 3042- Assignment 3- Winter 2026- Solutions

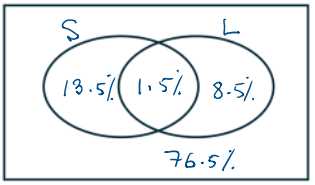
February 4, 2026 2:25 PM

Problem 1: (3 marks) A compiler encounters syntax errors with probability 0.15 (15%) and logic errors with probability 0.10. If these are independent, what's the probability of:

a) Both types of error? $0.15 \times 0.10 = 0.015 = 1.5\%$

b) At least one type of error? 23.5%

c) No error? 76.5%



$100\% - 23.5\% = 76.5\%$

Problem 2: (5 marks) An authentication system has:

- 1% of login attempts are hackers
- System blocks 98% of hacker attempts $0.98 \times 1\% = 0.98\%$
- System incorrectly blocks 2% of legitimate attempts. $0.99 \times 2\% = 1.98\%$

a) If a login is blocked, what's the probability it was actually a hacker?

b) What percent of login attempts are blocked by the system?

	Hacker	Legitimate	Sum
Block	0.98%	1.98%	2.96%
Does not Block	0.02%	97.02	97.04%
	1%	99%	100%

a) $\frac{0.98\%}{2.96\%} = 33.1\%$

b) 2.96%

Problem 3: (4 marks) Let X = number of bugs found in code review (0 to 5 bugs), with

$P(X = k) = \frac{6-k}{21}$ $for\ k\ =\ 0,1,2,3,4,5.$

Find E[X] and Var(X).

x	$P(x = k)$	$x.P(x)$	$x^2P(x)$
0	6/21	0	0
1	5/21	5/21	5/21
2	4/21	8/21	16/21
3	3/21	9/21	27/21
4	2/21	8/21	32/21
5	1/21	5/21	25/21
SUM	1	$\frac{35}{21}$	$\frac{105}{21}$

$E[X] = \frac{35}{21} \simeq 1.7$ bugs

$Var = \frac{105}{21} - \left(\frac{35}{21}\right)^2 = \frac{20}{9} \simeq 2.2$

$stdev = \sqrt{2.2} \simeq 1.5$

Average number of bugs : $\boxed{1.7 \pm 1.5}$

Problem 5: (3 marks) A system password must contain at least one digit. There are 62 characters (26 lowercase + 26 uppercase + 10 digits). For an 8-character password calculate:

P(at least one digit):

$P = \frac{62^8 - 52^8}{62^8} = \boxed{75.5\%}$

All : 62^8

no digits : 52^8

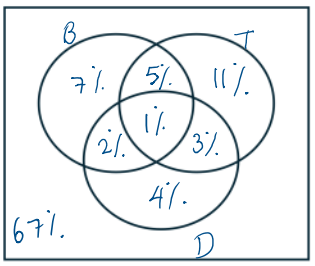
At least one digit : $62^8 - 52^8$

Problem 6: (5 marks) A CI/CD pipeline can fail at build stage (B), test stage (T), or deployment stage (D):

Given that: P(B) = 0.15, P(T) = 0.20 , P(D) = 0.10 , P(B ∩ T) = 0.06 (both build and test fail), P(B ∩ D) = 0.03 , P(T ∩ D) = 0.04 , P(B ∩ T ∩ D) = 0.01

Determine the P(pipeline succeeds) = 67%

Hint: Use the inclusion-exclusion principle to find P(B ∪ T ∪ D) first, then use the complement rule.

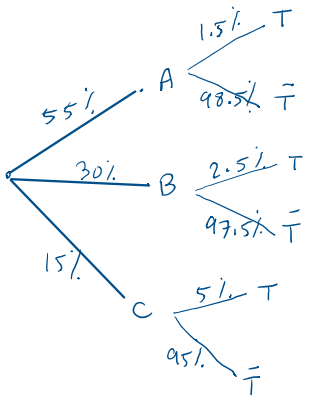


100% - 33% = 67%

Problem 7: (5 marks) An AI model inference system uses 3 GPU clusters:

- 55% of requests go to Cluster A, where 1.5% timeout (T)
- 30% of requests go to Cluster B, where 2.5% timeout
- 15% of requests go to Cluster C, where 5% timeout

Calculate P(timeout):



$(0.55 \times 0.015) + (0.30 \times 0.025) + (0.15 \times 0.05)$

$= 0.02325 \simeq \boxed{2.3\%}$ overall probability of time out.

Problem 8: (5 marks) SQL Injection Detection: Given:

- P(malicious query) = 0.005 (0.5% of queries are malicious)
- P(WAF flags | malicious) = 0.92 (true positive rate) $0.92 \times 0.5\% = 0.46\%$
- P(WAF flags | legitimate) = 0.03 (false positive rate) Find: $0.03 \times 99.5\% = 2.985\%$

a) P(malicious | WAF flags) $= \frac{0.46}{3.445} = 13.35\%$

b) What percent of flagged queries are actually malicious? Same

	Malicious	Legitimate	Sum
Flag	0.46%	2.985%	3.445%
No Flag	0.04%	96.515%	96.555%
	0.5%	99.5%	100%